CLASS 9 — MATHS · FORMULA SHEET

Chapter 1: **Coordinates** | Quick revision, one page

Cartesian plane

- Two perpendicular number lines meet at the origin $O(0, 0)$: the horizontal x-axis and the vertical y-axis.
- A point is written as an ordered pair (x, y): x is the abscissa (distance along the x-axis) and y is the ordinate.
- Point on the x-axis: $(x, 0)$. Point on the y-axis: $(0, y)$. Axis points lie in no quadrant.

Quadrants

- Quadrant I: $(+, +)$. Quadrant II: $(-, +)$. Quadrant III: $(-, -)$. Quadrant IV: $(+, -)$.
- Quadrants are numbered anticlockwise starting from the top right.

Reflections

- In the x-axis: $(x, y) \to (x, -y)$.
- In the y-axis: $(x, y) \to (-x, y)$.
- In both axes (through the origin): $(x, y) \to (-x, -y)$.

Distance and midpoint

- Distance between $A(x_1, y_1)$ and $B(x_2, y_2)$: $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.
- Distance of (x, y) from the origin: $\sqrt{x^2 + y^2}$.
- Points with the same y: distance $= |x_2 - x_1|$. Points with the same x: distance $= |y_2 - y_1|$.
- Midpoint of AB: $\left(\dfrac{x_1 + x_2}{2}, \dfrac{y_1 + y_2}{2} \right)$.

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